

1 Multi-allele model

Def. • Locus = specific fixed position on a chromosome where a particular gene is located

• Allele = different variant forms of a gene that can exist at a single locus

Haploid population of size N (each individual has only one allele for a given locus, on 1 chromosome)

Locus with $s+1$ possible alleles: $A_1, \dots, A_{s+1}$

Initially: $P(A_i) = x_i \Rightarrow P(A_{s+1}) = x_{s+1} = 1 - \sum_{i=1}^s x_i$

1 generation later: # A_i alleles in the population = m_i
 $\Rightarrow \sum_{i=1}^{s+1} m_i = N$

When the N individuals of the next generation are treated as N independent draws (random sampling with replacement) from the parent generation's gene pool, where the probability of drawing allele A_i is its current frequency, x_i , we get the multi-nomial

distribution:

$$P(\{m_i\}) = \binom{N}{m_1, \dots, m_{s+1}} \cdot \prod_{i=1}^{s+1} (P(A_i))^{m_i} = \frac{N!}{m_1! \dots m_{s+1}!} \prod_{i=1}^{s+1} x_i^{m_i}$$

Note: after one allele is drawn to become the gene of one offspring we put it back in the urn. "replacement" ensures that $\text{Prob}(\text{drawing } A_i)$ remains constant at x_i for each of the N independent draws forming the next generation.

1.a)*) $\Delta x_i = \frac{m_i}{N} - x_i$, since $E(m_i) = Nx_i$:

$$E(\Delta x_i) = \frac{Nx_i}{N} - x_i = 0 = \langle \Delta x_i \rangle$$

*) $\Delta x_i^2 = \left(\frac{m_i}{N} - x_i\right)^2 = \frac{m_i^2}{N^2} + x_i^2 - \frac{2x_i m_i}{N}$

$$E(\Delta x_i^2) = \frac{E(m_i^2)}{N^2} + x_i^2 - \frac{2x_i}{N} \cdot E(m_i)$$

$$\text{Var}(m_i) = E(m_i^2) - [E(m_i)]^2$$

$$\rightarrow E(\Delta x_i^2) = \frac{\text{Var}(m_i) + [E(m_i)]^2}{N^2} + x_i^2 - 2x_i^2$$

Let $y_k^i = \begin{cases} 1 & \text{if } k^{\text{th}} \text{ individual receives allele } A_i \\ 0 & \text{otherwise} \end{cases}$

$$\Rightarrow \text{Var}(m_i) = \text{Var}\left(\sum_{k=1}^N y_k^i\right) = \sum_{k=1}^N \text{Var}(y_k^i)$$

Random sampling in Wright-Fisher implies $\text{Cov}(y_k^i, y_l^i) = 0$ for $k \neq l$

y_k^i is a Bernoulli RV

$$\Rightarrow \text{Var}(y_k^i) = P(y_k^i = 1) \cdot P(y_k^i = 0) = x_i(1-x_i)$$

$$\hookrightarrow \text{Var}(m_i) = N x_i(1-x_i)$$

$$\begin{aligned} \hookrightarrow E(\Delta x_i^2) &= \frac{N x_i(1-x_i) + [N x_i]^2}{N^2} - x_i^2 \\ &= \frac{N x_i - N x_i^2 + N^2 x_i^2 - N^2 x_i^2}{N^2} = \frac{x_i(1-x_i)}{N} \end{aligned}$$

$$\rightarrow \langle \Delta x_i^2 \rangle = x_i(1-x_i)/N$$

$$*) \Delta x_i \Delta x_j = \left(\frac{m_i}{N} - x_i\right)\left(\frac{m_j}{N} - x_j\right) = \frac{1}{N^2} m_i m_j - \frac{1}{N} (m_i x_j + x_i m_j) + x_i x_j$$

$$\begin{aligned} E(\Delta x_i \Delta x_j) &= \frac{1}{N^2} E(m_i m_j) - \frac{1}{N} (N x_i \cdot x_j + x_i \cdot N x_j) + x_i x_j \\ &= \frac{E(m_i m_j)}{N^2} - x_i x_j \end{aligned}$$

$$m_i = \sum_{k=1}^N y_k^i \quad \text{and} \quad m_j = \sum_{l=1}^N y_l^j :$$

$$\begin{aligned} E(m_i m_j) &= E\left(\sum_{k,l=1}^N y_k^i y_l^j\right) \\ &= \sum_{k=1}^N \underbrace{E(y_k^i y_k^j)}_0 + 2 \sum_{1 \leq k < l \leq N} \underbrace{E(y_k^i y_l^j)}_{E(y_k^i) E(y_l^j) = x_i x_j} \end{aligned}$$

$y_k^i y_k^j$ is a indicator variable
equalling 1 if k receives both
alleles A_i & A_j , which is impossible
in a haploid population

E factorises from independent
random sampling assumption
in the W-F model. We have
 $\frac{(N-1)N}{2}$ identical terms

$$\Rightarrow E(\Delta x_i \Delta x_j) = \frac{2 \cdot \frac{(N-1)N}{2} \cdot x_i x_j}{N^2} - x_i x_j = \frac{(N-1) x_i x_j}{N} - x_i x_j$$

$$\langle \Delta x_i \Delta x_j \rangle = -\frac{x_i x_j}{N} \quad i \neq j$$

1.b) The general multi-variate Fokker-Planck equation is given by:

$$\frac{\partial \rho(\vec{x}, t)}{\partial t} = - \sum_{i=1}^s \frac{\partial}{\partial x_i} [\underbrace{v_i(\vec{x})}_{\text{drift}} \rho(\vec{x}, t)] + \sum_{i,j=1}^s \frac{\partial^2}{\partial x_i \partial x_j} [\underbrace{D_{ij}(\vec{x})}_{\text{diffusion}} \rho(\vec{x}, t)]$$

where $\boxed{v_i(\vec{x}) = \frac{\langle \Delta x_i \rangle}{\Delta t}}$ and $\boxed{D_{ij}(\vec{x}) = \frac{1}{2} \frac{\langle \Delta x_i \Delta x_j \rangle}{\Delta t}}$

are generalisations of the 1D case $v(x) = \frac{\langle \Delta x \rangle}{\Delta t}$, $D(x) = \frac{1}{2} \frac{\langle \Delta x^2 \rangle}{\Delta t}$

From 1.a) . $\langle \Delta x_i \rangle = 0 \Rightarrow v_i(\vec{x}) = 0$

$$\begin{cases} \langle \Delta x_i^2 \rangle = \frac{x_i(1-x_i)}{N} \\ \langle \Delta x_i \Delta x_j \rangle = -\frac{x_i x_j}{N} \end{cases} \Rightarrow D_{ij}(\vec{x}) = \frac{1}{2N} [x_i \delta_{ij} - x_i x_j]$$

We are left with:

$$\boxed{\frac{\partial \rho(\vec{x}, t)}{\partial t} = \frac{1}{2N} \sum_{i,j=1}^s \frac{\partial^2}{\partial x_i \partial x_j} [(x_i \delta_{ij} - x_i x_j) \rho(\vec{x}, t)]}$$

1.c) General form of the multivariate backward Kolmogorov equation, describing the evolution of probability $\rho(\vec{x}, t | \vec{y})$ of reaching final state $\vec{x}$ at time t from initial state $\vec{y} = (y_1, \dots, y_s)$ of allele frequencies:

$$\frac{\partial \rho(\vec{x}, t | \vec{y})}{\partial t} = \sum_{i=1}^s V_i(\vec{y}) \frac{\partial \rho}{\partial y_i} + \sum_{i,j=1}^s D_{ij}(\vec{y}) \frac{\partial^2 \rho}{\partial y_i \partial y_j}$$

Again, from 1.a) we are left with:

$$\frac{\partial \rho(\vec{x}, t | \vec{y})}{\partial t} = \frac{1}{2N} \sum_{i,j=1}^s (y_i \delta_{ij} - y_i y_j) \frac{\partial^2 \rho(\vec{x}, t | \vec{y})}{\partial y_i \partial y_j}$$

1.d) We first define the probability that the polymorphism still exists at time t (hasn't been absorbed yet).

Probability that population is still polymorphic is denoted as the survival probability:

$$S(t | \vec{y}) = \int_{\hat{A}} d^s \vec{x} \rho(\vec{x}, t | \vec{y}) = \mathbb{P}(\text{still polymorphic at } t | \vec{y})$$

where $\hat{A}$ is the interior region of the allele probability simplex, where every allele is still present:

$$x_i > 0 \text{ for } i = 1, \dots, s+1 \text{ and } \sum_{i=1}^{s+1} x_i = 1$$

The integral is thus over the non-absorbing states.

Notice, if we take T to be the random time at which the polymorphism disappears (i.e. population becomes monomorphic with $y_i = 1, y_{j \neq i} = 0$)

the CDF for T is: $F(t) = \mathbb{P}(T \leq t | \vec{y}) = 1 - S(t | \vec{y})$

$$\rightarrow p_x(t | \vec{y}) = \frac{d}{dt} F(t) = \frac{d}{dt} \int_0^t p_x(u | \vec{y}) du = - \frac{d}{dt} S(t | \vec{y})$$

$$\Rightarrow p_x(t | \vec{y}) = - \frac{d}{dt} \int_{V_0} d^s \vec{x} \rho(\vec{x}, t | \vec{y})$$

$$p_x(t | \vec{y}) = - \int_{V_0} d^s \vec{x} \frac{d\rho(\vec{x}, t | \vec{y})}{dt}$$

Mean fixation time:

$$\langle T(\vec{y}) \rangle_x = \int_0^\infty t \cdot p_x(t | \vec{y}) dt = - \int_0^\infty t \cdot \frac{d}{dt} S(t | \vec{y}) dt$$

$$\stackrel{\text{T.B.P.}}{=} \underbrace{\left[-t \cdot S(t | \vec{y}) \right]_{t=0}^{t=\infty}}_{\text{0}} + \int_0^\infty S(t | \vec{y}) dt$$

0, assuming $S(t | \vec{y}) \xrightarrow{t \rightarrow \infty} 0$ sufficiently fast

Defining $B_{\vec{y}} = \frac{1}{2N} \sum_{i,j=1}^S y_i (\delta_{ij} - y_j) \frac{\partial^2}{\partial y_i \partial y_j}$:

$$B_{\vec{y}} \langle \tau(\vec{y}) \rangle_x = \int_0^\infty B_{\vec{y}} S(t|\vec{y}) dt \stackrel{\text{swapped integration order}}{=} \int_{\mathcal{R}} d^S \vec{x} \int_0^\infty \underline{B_{\vec{y}} \rho(\vec{x}, t|\vec{y})} dt$$

From 1.c) we have $B_{\vec{y}} \rho(\vec{x}, t|\vec{y}) = \partial_t \rho(\vec{x}, t|\vec{y})$:

$$\begin{aligned} \Rightarrow B_{\vec{y}} \rho(\vec{x}, t|\vec{y}) &= \int_{\mathcal{R}} d^S \vec{x} \int_0^\infty \frac{\partial \rho(\vec{x}, t|\vec{y})}{\partial t} dt \\ &= \int_{\mathcal{R}} d^S \vec{x} \left[\underbrace{\rho(\vec{x}, \infty|\vec{y})}_0 - \underbrace{\rho(\vec{x}, 0|\vec{y})}_{\delta(\vec{x}-\vec{y})} \right] \end{aligned}$$

since $\vec{x}$ are non-absorbing states,
but at ∞ we necessarily become monomorphic

Since $\vec{y} \in \mathcal{R}$ (polymorphic state) :

$$\boxed{\frac{1}{2N} \sum_{i,j=1}^S y_i (\delta_{ij} - y_j) \frac{\partial^2 \langle \tau(\vec{y}) \rangle_x}{\partial y_i \partial y_j} = -1}$$

For two alleles: $S=1$, $y_1=y$, $y_2=1-y$:

$$\frac{1}{2N} y(1-y) \frac{\partial^2}{\partial y^2} \langle f(y) \rangle_x = -f$$

matching eq. (1.89) from the MIT notes

2 Splitting probability and mean first passage time:

2.a) $v=0$ $D=\text{const}$

Initially, particle starts at $0 < y < 1$

Diffusion until absorption at either boundary

$x_0=0$ or $x_1=1$.

$$\Pi^*(x_a, y) = \lim_{t \rightarrow \infty} p(x_a, t | y) = \Pi_a(y)$$

= Prob. of starting at y and fixing at absorbing state x_a after a long time

$\Pi_a(y) = \Pi^*(a, y)$; $a = \{0, 1\}$, are steady state solutions of the backward Kolmogorov eqn°.

$$D \frac{d^2 \Pi_a}{dy^2} = 0 \Rightarrow \frac{d \Pi_a}{dy} = C_1 \Rightarrow \Pi_a(y) = C_1^a y + C_2^a$$

Boundary conditions:

absorption at $x_0 = 0$: $\Pi_0(0) = 1$; $\Pi_0(1) = 0$

absorption at $x_1 = 1$: $\Pi_1(1) = 1$; $\Pi_1(0) = 0$

Solving steady state:

at $x_0 = 0$:

$$\begin{cases} \Pi_0(0) = C_2^0 = 1 \\ \Pi_0(1) = C_1^0 + 1 = 0 \end{cases} \Rightarrow \boxed{\Pi_0(y) = -y + 1}$$

at $x_1 = 1$:

$$\begin{cases} \Pi_1(0) = C_2^1 = 0 \\ \Pi_1(1) = C_1^1 = 1 \end{cases} \Rightarrow \boxed{\Pi_1(y) = y}$$

Sanity check: $\Pi_0(y) + \Pi_1(y) = 1 \quad \checkmark$

2. b) Def.: First passage time := how long it takes for the particle to reach any absorbing boundary.

• $\langle \tau(y) \rangle_x$ (MIT notes eq. 1.87) = average total time the system remains in the polymorphic, non-absorbing state region. (eq. 1.88, MIT)

$$\hookrightarrow \bar{E}(y) \equiv \langle \tau(y) \rangle_x \Rightarrow \text{By } \bar{E}(y) = -1$$

$$\rightarrow \frac{d^2 \bar{E}(y)}{dy^2} = -1, \text{ with B.C. : } \bar{E}(0) = 0 = \bar{E}(1)$$

$$\hookrightarrow \frac{d^2 \bar{E}(y)}{dy^2} = -\frac{1}{0} \Rightarrow \frac{d\bar{E}(y)}{dy} = -\frac{y}{0} + C_1$$

$$\Rightarrow \bar{E}(y) = -\frac{y^2}{20} + C_1 y + C_2$$

Apply B.C. : $\begin{cases} \bar{E}(0) = C_2 = 0 \\ \bar{E}(1) = -\frac{1}{20} + C_1 = 0 \end{cases} \Rightarrow \bar{E}(y) = \frac{y}{20}(1-y)$

maximising $\bar{E}(y)$ by setting derivative to zero:

$$\left. \frac{d\bar{E}(y)}{dy} \right|_{y_{\max}} = \frac{1}{20}(1 - 2y_{\max}) = 0 \Rightarrow y_{\max}^{\text{rd}} = \frac{1}{2}$$

2.c) We already noted that the fixation probabilities are steady state solutions of BKE. Going to 2D we have:

$$\nabla^2 \Pi_a = 0 \xrightarrow{\text{Polar Coordinates}} \frac{d^2}{dR^2} \Pi_a(R) + \frac{1}{R} \frac{d}{dR} \Pi_a(R) = 0$$

$$\text{Let } u = \frac{d\Pi_a}{dR}: \quad \frac{du}{dR} = -\frac{1}{R} u \Rightarrow \frac{du}{u} = -\frac{dR}{R}$$

$$\Rightarrow \ln(u) = -\ln(R) + \ln(C_1) = \ln\left(\frac{C_1}{R}\right) \quad (A)$$

$$\Rightarrow \frac{d\Pi_a(R)}{dR} = u = \frac{C_1}{R} \Rightarrow \Pi_a(R) = C_1^a \ln(R) + C_2^a$$

B.C.s:

Inner wall absorption: $\Pi_0(R_0) = 1$; $\Pi_0(1) = 0$

Outer wall absorption: $\Pi_1(R_1) = 1$; $\Pi_1(0) = 0$

$$\begin{cases} \Pi_0(R_0) = C_1^0 \ln(R_0) + C_2^0 = 1 & (\beta 1) \end{cases}$$

$$*) \begin{cases} \Pi_0(R_1) = C_1^0 \ln(R_1) + C_2^0 = 0 & (\beta 2) \end{cases}$$

$$\stackrel{(B1)}{\Rightarrow} C_2^0 = -C_1^0 \ln(R_1) \stackrel{(B2)}{\Rightarrow} C_1^0 \ln(R_0/R_1) = 1$$

$$\stackrel{(A)}{\hookrightarrow} \Pi_0(R) = \frac{1}{\ln\left(\frac{R_0}{R_1}\right)} \left[\ln(R) - \ln(R_1) \right]$$

$$\Pi_0^{2-d}(R) = \frac{\ln(R/R_1)}{\ln(R_0/R_1)}$$

$$**) \begin{cases} \Pi_1(R_0) = C_1^1 \ln(R_0) + C_2^1 = 0 & (b1) \\ \Pi_1(R_1) = C_1^1 \ln(R_1) + C_2^1 = 1 & (b2) \end{cases}$$

$$\stackrel{(b1)}{\Rightarrow} C_2^1 = -C_1^1 \ln(R_0) \stackrel{(b2)}{\Rightarrow} C_1^1 \ln(R_1/R_0) = 1$$

$$\stackrel{(A)}{\hookrightarrow} \Pi_1(R) = \frac{1}{\ln\left(\frac{R_1}{R_0}\right)} \left[\ln(R) - \ln(R_0) \right]$$

$$\Pi_1^{2-d}(R) = \frac{\ln(R/R_0)}{\ln(R_1/R_0)} = 1 - \Pi_0(R)$$

Mean fixation time: $\nabla^2 \bar{E}(R) = -1$

$$\begin{aligned} & \frac{d^2 \bar{E}}{dR^2} + \frac{1}{R} \frac{d\bar{E}}{dR} = -\frac{1}{D} \\ (xR) \quad & \rightarrow R \frac{d^2 \bar{E}}{dR^2} + \frac{d\bar{E}}{dR} = -\frac{R}{D} \end{aligned}$$

$$\Leftrightarrow \frac{d}{dR} \left(R \frac{d\bar{E}}{dR} \right) = -\frac{R}{D} \Rightarrow R \frac{d\bar{E}}{dR} = -\frac{R^2}{2D} + C_3$$

$$\Leftrightarrow \frac{d\bar{E}}{dR} = -\frac{R}{2D} + \frac{C_3}{R} \Rightarrow \boxed{\bar{E}(R) = -\frac{R^2}{4D} + C_3 \ln(R) + C_4}$$

B.C.: $\bar{E}(R_0) = 0 = \bar{E}(R_1)$

$$\hookrightarrow \bar{E}(R_0) = -\frac{R_0^2}{4D} + C_3 \ln(R_0) + C_4 = 0 \quad (l_1)$$

$$\bar{E}(R_1) = -\frac{R_1^2}{4D} + C_3 \ln(R_1) + C_4 = 0 \quad (l_2)$$

$$\cdot (l_1) - (l_2) \Rightarrow \frac{R_1^2 - R_0^2}{4D} + C_3 \ln\left(\frac{R_0}{R_1}\right) = 0$$

$$\hookrightarrow C_3 = \frac{R_1^2 - R_0^2}{4D \ln(R_1/R_0)}$$

Substitution C_3 back in (2):

$$C_4 = \frac{R_1^2}{4D} - \frac{(R_1^2 - R_0^2)}{4D \ln(R_1/R_0)} \ln(R_1)$$

$$\rightarrow \bar{E}^{2d}(R) = \frac{1}{4D} \left[(R_1^2 - R^2) - \frac{(R_1^2 - R_0^2)}{4D \ln(R_1/R_0)} \ln(R_1/R) \right]$$

$$\left. \frac{d\bar{E}(R)}{dR} \right|_{R_{\max}} = -\frac{R_{\max}}{2D} + \frac{R_1^2 - R_0^2}{4D \ln(R_1/R_0) R_{\max}} = 0$$

$$\Rightarrow R_{\max}^2 = \frac{R_1^2 - R_0^2}{2 \ln(R_1/R_0)} \Rightarrow R_{\max}^{2d} = + \sqrt{\frac{R_1^2 - R_0^2}{2 \ln(R_1/R_0)}}$$

2.d) 3D Laplacian:

$$\nabla^2 = \frac{1}{R^2} \frac{d}{dR} \left(R^2 \frac{d}{dR} \right) = \frac{1}{R^2} \left[2R \frac{d}{dR} + R^2 \frac{d^2}{dR^2} \right] = \frac{2}{R} \frac{d}{dR} + \frac{d^2}{dR^2}$$

$$\nabla^2 \Pi_a(R) = 0 \Rightarrow \frac{d^2 \Pi_a}{dR^2} = -\frac{2}{R} \frac{d\Pi_a}{dR}$$

Similar to before:

$$\ln\left(\frac{d\Pi_a}{dR}\right) = -2 \ln(R) + \ln C = \ln\left(\frac{C}{R^2}\right)$$

$$\frac{d\Pi_a}{dR} = \frac{C}{R^2} \Rightarrow \boxed{\Pi_a(R) = \frac{C}{R} + \tilde{C}}$$

B.C for inner absorption: $\Pi_0(R_0) = 1$; $\Pi_0(R_1) = 0$

$$\begin{cases} \Pi_0(R_0) = \frac{C}{R_0} + \tilde{C} = 1 \\ \Pi_0(R_1) = \frac{C}{R_1} + \tilde{C} = 0 \end{cases} \rightarrow \tilde{C} = -\frac{C}{R_1}$$

$$\hookrightarrow 1 = C\left(\frac{1}{R_0} - \frac{1}{R_1}\right)$$

$$\Pi_0(R) = \frac{C}{R} - \frac{C}{R_1} = \frac{R_0 R_1}{R_1 - R_0} \left(\frac{R_1 - R}{R R_1} \right)$$

$$\boxed{\overset{3-d}{\Pi_0(R)} = \frac{R_0(R_1 - R)}{R(R_1 - R_0)}};$$

$$\Pi_1(R) = 1 - \Pi_0 = \frac{R R_1 - \cancel{R R_0} - R_0 R_1 + \cancel{R_0 R}}{R(R_1 - R_0)} = \frac{R_1(R - R_0)}{R(R_1 - R_0)}$$

$$\Pi_1(R) = \frac{R_1(R_0 - R)}{R(R_0 - R_1)}$$

Mean exit time: $\nabla^2 \bar{E} = -1$, in 3D we get:

$$\frac{d^2 \bar{E}}{dR^2} + \frac{2}{R} \frac{d\bar{E}}{dR} = -\frac{1}{D}$$

Multiplying by R^2 (integrating factor):

$$\frac{d}{dR} \left(R^2 \frac{d\bar{E}}{dR} \right) = -\frac{R^2}{D} \Rightarrow \frac{d\bar{E}}{dR} = -\frac{R}{3D} + \frac{C_5}{R^2}$$

$$\Rightarrow \boxed{\bar{E}(R) = -\frac{R^2}{6D} - \frac{C_5}{R} + C_6}$$

using $\bar{E}(R_0) = 0 = \bar{E}(R_1)$

$$\begin{cases} -\frac{R_0^2}{6D} - \frac{C_5}{R_0} + C_6 = 0 \\ -\frac{R_1^2}{6D} - \frac{C_5}{R_1} + C_6 = 0 \end{cases}$$

$$C_5 = -\frac{(R_1^2 - R_0^2)}{6D \left(\frac{1}{R_1} - \frac{1}{R_0} \right)} = -\frac{R_1^2 - R_0^2}{6D} \frac{R_1 R_0}{(R_0 - R_1)} = -\frac{R_0 R_1 (R_1 + R_0)}{6D}$$

$$C_6 = \frac{C_5}{R_1} + \frac{R_1^2}{6D}$$

$$\hookrightarrow \bar{E}(R) = \frac{1}{6D} \left[(R_1^2 - R^2) + C_5 \left(\frac{1}{R_1} - \frac{1}{R} \right) \right]$$

We get:

$$\bar{E}(R) = \frac{1}{6D} \left[(R_1^2 - R^2) - (R_1^2 - R_0^2) \frac{R_0(R_1 - R)}{R(R_1 - R_0)} \right]$$

$$\begin{aligned} \frac{d\bar{E}(R)}{dR} \Big|_{R_{\max}} &= -\frac{R_{\max}}{3D} + (R_1 + R_0) R_0 \frac{d}{dR} \left[\frac{R_1}{R} - 1 \right] \Big|_{R_{\max}} \\ &= -\frac{R_{\max}}{3D} - \frac{(R_1 + R_0) R_0 R_1}{6D R_{\max}^2} = 0 \end{aligned}$$

$\Rightarrow$

$$R_{\max} = \left(\frac{R_0 R_1 (R_1 + R_0)}{2} \right)^{1/3}$$

3 Number of gapped alignments:

3. a) Suppose we have two ordered sequences of symbols:

$$S_A = (A_1, \dots, A_n)$$
$$S_B = (B_1, \dots, B_m)$$

Let $f(n, m)$ be the # of ways to intercalate A & B into a single sequence, S_B^A , of length $n+m$ while preserving the internal order of each sequence.

Recursive proof:

if $n=0$: $f(0, m) = 1$

if $m=0$: $f(n, 0) = 1$

Notice there are only 2 possible choices for the last element of the intercalated sequence:

$$S_B^A[n+m] = \begin{cases} A_n, & \text{or} \\ B_m \end{cases}$$

$S_B^A[n+m] = A_n \rightarrow f(n-1, m)$ ways of having S_B^A end with A_n

$S_B^A[n+m] = Bm \rightarrow f(n, m-1)$ ways of having
 S_B^A end with Bm

$$\Rightarrow f(n, m) = f(n-1, m) + f(n, m-1)$$

We notice that this is the same recursion and boundary conditions from Pascal's rule if we identify $f(n, m) = \binom{n+m}{n}$:

$$\Rightarrow \binom{n+m}{n} = \binom{n-1+m}{n-1} + \binom{n+m-1}{n}$$

Indeed:

$$\begin{aligned} \frac{(n-1+m)!}{(n-1)! m!} + \frac{(n+m-1)!}{n! (m-1)!} &= \frac{n(n-1+m)! + m(n+m-1)!}{n! m!} \\ &= \frac{(n+m-1)! (n+m)}{n! m!} \\ &= \frac{(n+m)!}{n! (n+m-n)!} \end{aligned}$$

3.b) Proof of one-to-one correspondence between gapped alignments of two sequences and intercalated sequences.

Let $S_A = (A_1, \dots, A_n)$ and $S_B = (B_1, \dots, B_m)$ be two sequences

Set G := all valid gapped global alignments of S_A & S_B .

Set I := all intercalations, S_B^A , preserving symbol order of subsequences S_A and S_B

We construct explicit mappings: $\Phi: G \mapsto I$
 $\Psi: I \mapsto G$

Forward mapping: $G \ni G \rightarrow I \in I$

$G = \left[\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \dots, \begin{pmatrix} x_L \\ y_L \end{pmatrix} \right] = [c_1, \dots, c_L]$, where each c_k is either:

$\begin{cases} (A_i, -) & , \text{gap in } B \\ (-, B_j) & , \text{gap in } A \\ (A_i, B_j) & , \text{aligned pair} \end{cases}$

such that each A_i , $i = 1, \dots, n$ and B_j , $j = 1, \dots, m$ appears only once.

canonical ordering rule: with any alignment column, c_k , symbols are read upper row first, then lower row.

Algorithm for $\Phi(G) = I$:

Initialise empty set: $I \leftarrow \{\}$.

For $k=1$ to L :

if $c_k = (A_i, -)$: append A_i to I

if $c_k = (-, B_j)$: append B_j to I

if $c_k = (A_i, B_j)$: append A_i , then B_j to I

Return I

$$G = [c_1, \dots, c_L] \xrightarrow{\Phi} I = \{s_1, \dots, s_{n+m}\}$$

Reverse mapping: $I \ni I \rightarrow G \in \mathcal{G}$

We define i and j as pointers in S_A & S_B for the "next expected symbols", A_i, B_j .
 k is a pointer in the intercalated sequence S_A .

Algorithm for $\Psi(I) \rightarrow G$

Initialise pointers: $k \leftarrow 1, i \leftarrow 1, j \leftarrow 1$

Initialise empty alignment: $G \leftarrow []$

While $k \leq n + m$

if $(i \leq n) \wedge (j \leq m) \wedge (k < n + m) \wedge (s_k == A_i) \wedge (s_{k+1} == B_j)$:

Append (A_i, B_j) to G

$i \leftarrow i + 1, j \leftarrow j + 1, k \leftarrow k + 2$

else if $(i \leq n) \wedge (s_k == A_i)$:

Append $(A_i, -)$ to G

$i \leftarrow i + 1, k \leftarrow k + 1$

else if $(j \leq m) \wedge (s_k == B_j)$:

Append $(-, B_j)$ to G

$j \leftarrow j + 1, k \leftarrow k + 1$

else:

error, impossible for valid intercalation

Return G

where " $s_k = A_j$ " means that currently scanned symbol from S_B^A equals the expected symbol from A given pointer i which can only increment by 1 during the sweep of pointer k .

$$I = \{s_1, \dots, s_{n+m}\} \xrightarrow{\Psi} G = [c_1, \dots, c_L]$$

We clearly have $\Psi(\emptyset) = Id = \emptyset(\Psi)$

Thus the number of gapped alignments between two sequences of length n is:

$$\binom{2n}{n} = \frac{(2n)!}{n! n!} = \frac{(2n)!}{(n!)^2}$$

3. c) For large n we use Stirling's

formula: $n! \approx \sqrt{2\pi n} n^n e^{-n}$

$$\cdot (2n)! \approx \sqrt{4\pi n} (2n)^{2n} e^{-2n}$$

$$\cdot (n!)^2 = 2\pi n n^{2n} e^{-2n}$$

$$\rightarrow \binom{2n}{n} = \frac{(2n)!}{(n!)^2} \approx \frac{\sqrt{4\pi n} 2^{2n} \cancel{n^{2n}} \cancel{e^{-2n}}}{2\pi n \cancel{n^{2n}} \cancel{e^{-2n}}} = \frac{4^n}{\sqrt{\pi n}}$$

4 Alignments:

We want to find the optimal alignments for $A = (a_1, \dots, a_n)$ and $B = (b_1, \dots, b_m)$ using recursive algorithms for building score matrices.

$S_{i,j}$ = score of best alignment terminating on a_i, b_j
 $s(a,b)$ = score assigned to a match pair a,b

Arbitrary scoring convention used here:

$$\rightarrow \begin{cases} \text{Match} = +1.5 \\ \text{Mis-match} = -1 \end{cases}$$

$$\rightarrow \text{or } \downarrow : \text{Gap} = -2$$

- Global (Needleman - Wunsch) alignment

$$S_{ij} = \max \begin{cases} S_{i-1, j-1} + s(a_i, b_j) \\ S_{i-1, j} - g \\ S_{i, j-1} - g \end{cases}$$

where g is the linear gap penalty

Global alignment attempts to construct the single best match that spans both sequences.

		0	1	2	3	4	5
	.	.	G	A	T	T	A
0	.	0	-2	-4	-6	-8	-10
1	G	-2	1.5	-0.5	-2.5	-4.5	-6.5
2	A	-4	-0.5	3	1	-1	-3
3	A	-6	-2.5	1	2	0	0.5
4	T	-8	-4.5	-1	2.5	3.5	1.5
5	T	-10	-6.5	-3	0.5	4.0	2.5
6	C	-12	-8.5	-5	-1.5	2.0	3.0

← B.C. : $S(0, j) = -2j$

Traceback starts at final cell corresponding to full length of both sequences (stops at initial)

Global score = 3.0 = $S(6, 5)$

↑
B.C., $S(i, 0) = -2i$

We have two alternative tracebacks giving the same score $\Rightarrow$ Ambiguity

(3,2) via up :

S_1 :	G	A	A	T	T	C
S_2 :	G	A	-	T	T	A

Score : $+1.5 + 1.5 - 2 + 1.5 + 1.5 - 1 = 3$

(3,2) via diagonal :

S_1 :	G	A	A	T	T	C
S_2 :	G	-	A	T	T	A

Score : $+1.5 - 2 + 1.5 + 1.5 + 1.5 - 1 = 3$

• Local (Smith Waterman) alignment

$$S_{ij} = \max \begin{cases} 0 \\ S_{i-1,j-1} + s(a_i, b_j) \\ S_{i-1,j} - g \\ S_{i,j-1} - g \end{cases}$$

	.	G	A	T	T	A
.	0	0	0	0	0	0
G	0	1.5	0	0	0	0
A	0	0	3	1	0	1.5
A	0	0	1.5	2	0	1.5
T	0	0	0	3	3.5	1.5
T	0	0	0	1.5	4.5	2.5
C	0	0	0	0	2.5	3.5

Traceback starts from cell with maximum score found anywhere in the matrix. Stops as soon as it reaches any cell with value 0.

$\Rightarrow$ $S_1: A \ T \ T$, score : 4.5
 $S_2: A \ T \ T$

No ambiguity for local alignment, which identifies a shorter, higher scoring conserved region (score 4.5). Global alignment covers the full sequences with gaps, scoring (scoring 3).